Applied NLP

DATA 641

Lecture 15 — Machine Translation

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action, Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, First edition, MANNING

Machine translation

Machine Translation (MT) involves automatic translation of text from one language to another. It's one of the original challenges in NLP research. Approaches include:

Early Rule-Based Approaches:

- Early MT systems used rule-based approaches.
- Required extensive linguistic knowledge, including source and target language grammars.
- Necessitated explicit coding of linguistic resources like dictionaries.

Statistical Methods Era:

- Subsequent years saw the emergence of statistical methods.
- Depended on the availability of large parallel datasets between languages.
- Data often collected from sources like European parliamentary proceedings.

Rise of Deep Learning-Based Neural MT:

- In the past five years, deep learning-based neural MT approaches have gained prominence.
- These approaches have become the state of the art in both research and production-scale MT systems.
- Examples include Google Translate.
- Building and training deep learning-based MT systems require substantial data and resources. Typically led by large organizations due to resource requirements.

Usefulness of MT

Clearly, MT is a large research area, and building MT systems seems like a large effort. Where is MT useful in the industry? Here are two example scenarios where MT may be required to develop solutions:

- Our client's products are used by people around the world who leave reviews on social media in multiple languages. Our client wants to know the general sentiment of those reviews. For this, instead of looking for sentiment analysis tools in multiple languages, one option is to use an MT system, translate all the reviews into one language, and run sentiment analysis for that language.
- We work with a lot of social media data (e.g., tweets) on a regular basis and notice that it's unlike the kind of text we encounter in typical text documents. For example, consider the sentence, "am gud," which, in formal, well-formed English is, "I am good."MT can be used to map these two sentences by treating the conversion from "am gud" to "I am good" as an informal-to grammatical English translation problem.

Practical advices

Utilize Translation APIs:

- Prefer using translation APIs over building your own MT system when possible.
- Pay attention to pricing policies when using APIs.
- Consider storing translations of frequently used text in a translation memory or cache.

Translation Memory Maintenance:

- Maintain a translation memory for frequently recurring translations.
- Utilize the translation memory for improved efficiency.

Practical advices

Addressing New Languages or Domains:

- For new languages or domains where existing APIs perform poorly:
    - Consider domain knowledge-based, rule-based translation systems for specific scenarios.
    - Explore "back translation" to augment training data.

Hybrid MT Systems:

- For scenarios where translation accuracy is crucial:
    - Consider forming hybrid MT systems that combine neural models with rules and post-processing.
    - Hybrid systems can enhance translation quality.

MT Research and Evaluation:

- Machine translation is a broad research field with dedicated conferences, journals, and competitions.

- Academics and industry groups actively participate in MT research and system evaluation.

However, always keep in mind...

Translation is the art of failure

Attributed to Umberto Eco, an Italian novelist, philosopher etc. Eco was known for his works on language, communication, and translation, and he made this statement to emphasize the challenges and limitations of translation in capturing the full nuance and richness of the original text.